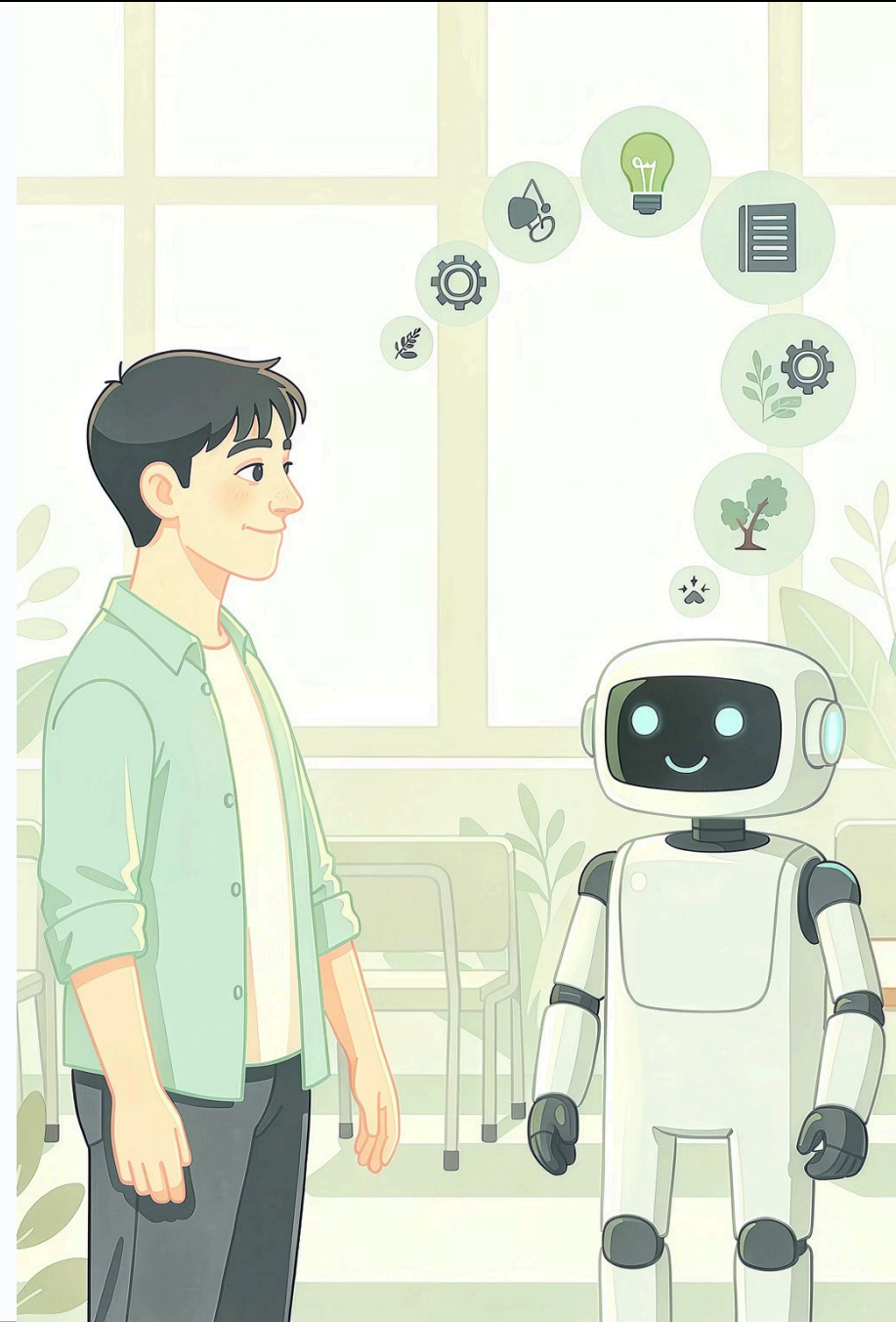


Session 11:

The Art of Prompting

Today, you'll learn what it means to be a **prompt engineer** - someone who knows how to give AI clear, thoughtful instructions to get the best results. It's a bit like being a coach: you give clear directions, explain what you want, and guide the AI to bring your ideas to life.

Whether you're working on projects, exploring topics you're curious about, or just having fun, this skill will help you get exactly what you need from AI.



Objectives

By the end of this session, you will be able to:

01

Define what a prompt is and why it matters

02

Identify key elements that can strengthen a prompt

03

Recognize when a prompt might be missing helpful details

04

Recognize common prompting mistakes and why they happen

05

Transform weak prompts into stronger ones

06

Use iterative prompting to improve AI responses

07

Apply The Prompt Framework

08

Start your Prompt Library

Recap: What We Learned in Session 10

Key Term	What It Means
Strength	What generative AI does well (speed, brainstorming, explaining).
Limit	What generative AI struggles with (no understanding, outdated info, bias).
Human-in-the-Loop	Keeping a human in control – AI assists, humans decide.
Appropriate Use	Using AI when it helps; avoiding AI when it harms or replaces thinking.

- ❏ Generative AI is powerful, but it has limits. Humans must stay in control, verify outputs, and decide when to use AI and when to avoid it. Today, we learn how to give AI instructions that respect those limits and get the best results.

Warm-Up

Have you ever asked a friend for help, and they did something completely different from what you meant? What went wrong?

Imagine you're a **coach** giving instructions to a team. You have a brilliant player who can do almost anything, but they need very clear directions. If you just say, *"Play,"* they'll stand there confused. But if you say, *"Run to the corner, catch the ball, and pass it to the player in the middle,"* they'll do exactly what you need.

AI is the same way. If you give it a vague instruction, it guesses what you want and sometimes guesses wrong.



What Is a Prompt?

A prompt is your instruction to AI.

A prompt is the words, questions, or commands you type to tell AI what you want. Think of AI like a very fast, very literal assistant. It can do amazing things, but it needs clear instructions.

Vague Prompt

"Make me something cool about snowman"

→ AI might give you a drawing of a snowman wearing sunglasses.

Clear Prompt

"Draw a detailed illustration of a penguin wearing a top hat on a beach at sunset"

→ You'll get something much closer to what you imagined.

Key Insight: AI doesn't read your mind. It only has your words. The clearer your words, the better your result. This is true for every AI tool you'll ever use – today, tomorrow, and in the future.

Think of it like ordering at a restaurant:

Weak order

"Food." → Confusion.

Better order

"A cheeseburger." → Still vague.

Excellent order

"I'd like a cheeseburger with extra cheese, lettuce, and tomato on a toasted bun, no pickles, with a side of sweet potato fries."

The clearer your order, the closer the result matches what you wanted.

Why Prompts Matter - Now and Later



GIGO Rule

Vague prompt → vague output.
Clear prompt → clear output.



Like Steering a Car

Your prompt is the steering wheel that guides the AI.



Human-in-the-Loop

You guide the AI. A good prompt is your first step in taking responsibility.



A Skill That Transfers

Works with ChatGPT, Gemini, NotebookLM, and future AI tools you haven't seen yet.



Better Answers

More relevant, accurate outputs you can trust.



Reduces Hallucinations

AI is less likely to guess wildly when your instructions are clear.

Key Elements of a Strong Prompt

A great prompt is built from several key elements. Each element adds a layer of clarity that helps the AI understand exactly what you want.

While there's no single "perfect" way to write a prompt, including certain elements often helps AI give you better answers. Think of these as **tools you can use when you need them** - not rules you must follow every time.



The Key Elements

Component	What It Means	Example
Clear Instructions	Tell the AI exactly what you want it to do. Use action words: <i>list, explain, create, summarize, compare.</i>	"Explain how volcanoes form."
Context	Give background: Who is the audience? What is the purpose?	"...for a 5th grader preparing a short presentation."
Format	Tell the AI how you want the answer to look: a list, a poem, a table, a paragraph, etc.	"Use 3 bullet points."
Constraints	Add limits: length, tone, style, what to include or avoid.	"Keep each bullet under 20 words. Don't use scientific names."

Putting the Elements Together

Weak Prompt

"Tell me about volcanoes"

Stronger Prompt (using elements)

Clear Instructions: Explain how volcanoes form

Context: for a 5th grader preparing a short presentation

Format: 3 bullet points

Constraints: each bullet under 20 words, no scientific names

Final Prompt

"Explain how volcanoes form for a 5th grader preparing a short presentation. Use 3 bullet points. Keep each bullet under 20 words. Don't use scientific names."

Important: These are *key* elements, but you can always add more details. For example, you might include a specific **tone** (funny, serious) or **audience** (my little sister) as part of Context or Constraints. The more you practice, the more you'll discover what works for your needs.

❗ **Pro Tip: Show, don't just tell.** Adding an **example** of what you want (e.g., "Write a description like this: ...") is one of the fastest ways to get perfect results. You can include an example as part of your Format or Constraints.

Mini Activity 1 : Spot the Missing Element(s)

Instructions: Look at each prompt below. Some are missing details that might help. Identify which of the key elements (Clear Instructions, Context, Format, Constraints) is missing, and explain why adding it could make the answer better. Remember: sometimes a short prompt is fine!

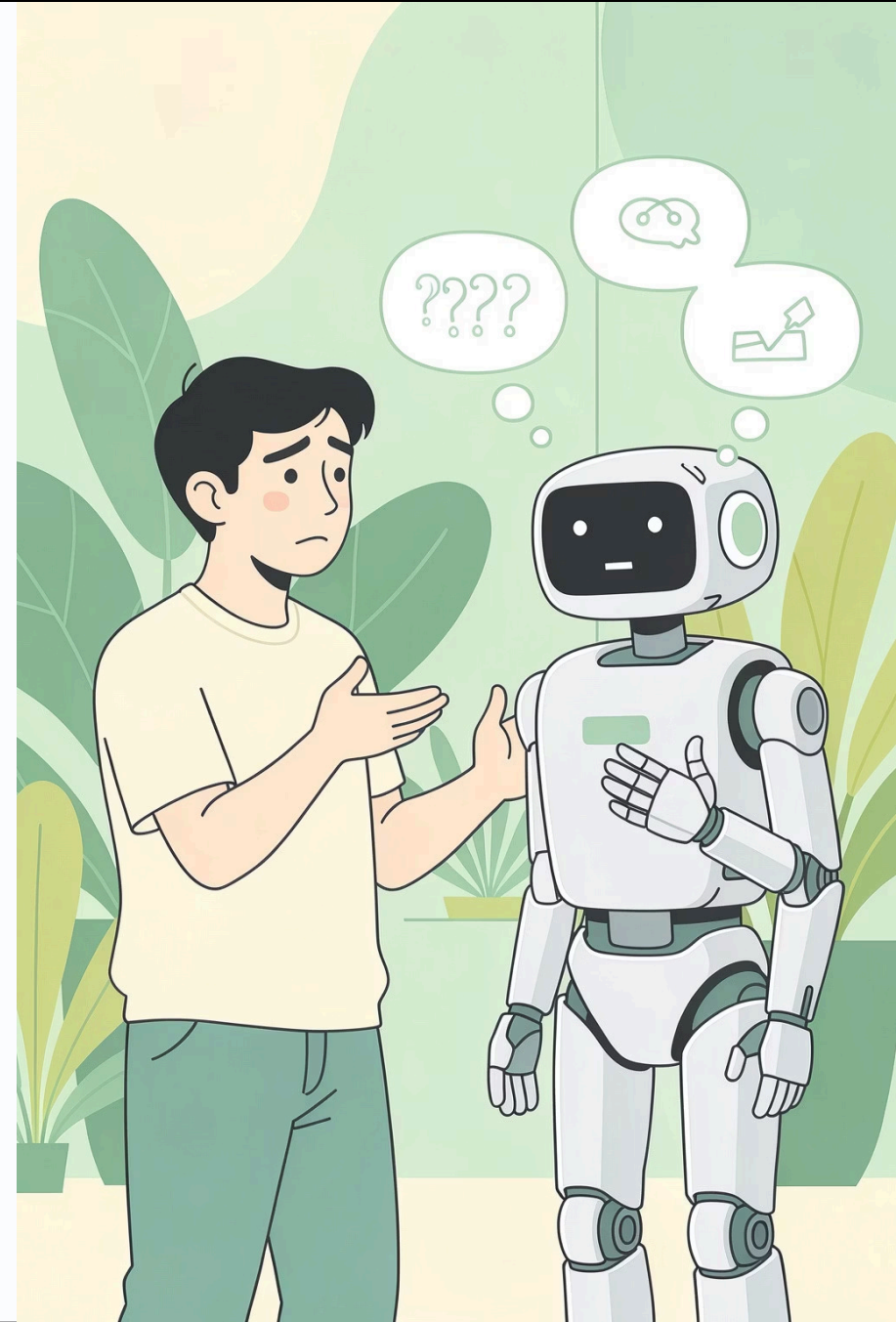
#	Prompt	Missing Element(s)	Why Adding It Might Help
1	"Tell me about Mars"		
2	"Explain how a car engine works"		
3	"Write a story"		
4	"Give me 5 ideas for a birthday party in a bulleted list"		
5	"Describe how to take care of a pet in simple words, no more than 3 sentences."		
6	"Help me plan a surprise party for my best friend"		

Mini Activity 1 - Answer Key

#	Missing Element(s)	Why Adding It Might Help
1	Context, Format, Constraints	Without context, AI doesn't know your age or interest; without format/constraints, answer may be too long or technical.
2	Context, Format, Constraints	Has instruction but no audience, no structure, no limits.
3	Context, Format, Constraints	Vague – no audience, no structure, no limits.
4	Context, Constraints	Has format (bulleted list) but no context about age or theme, and no constraints like length or style.
5	Instructions, Context	Has constraints and format, but no clear instruction (just "describe") and no context about audience.
6	Constraints	Has instruction, context, and format implied, but no constraints like length or style.

Common Prompting Mistakes

Even experienced users make these mistakes. Knowing them helps you avoid them. Each mistake has a real-world pattern - the way people actually use AI and an insight that helps you become a better prompt engineer.



Common Prompting Mistakes

❌ Assuming AI Understands You

People think AI "gets what they mean" without being specific.

Example: "Make this better."

Problem: AI doesn't know: better = shorter? funnier? clearer? more formal?

💡 AI does **not** understand intention - only **instructions**. You must translate your intention into explicit instructions.

❌ Giving Incomplete Information

Users forget context - they assume AI already knows.

Example: "Explain this topic."

Problem: What topic exactly? What level? For whom? This leads to: Generic answers, irrelevant outputs, wasted time revising.

💡 Context is not optional. Without it, the AI fills gaps with assumptions - which are often wrong.

❌ Not Checking Output Against Need

Users don't evaluate what AI gives them.

Example: AI gives an answer → user doesn't verify accuracy, relevance, usefulness.

💡 Always verify AI output against trusted sources, your own knowledge, and the actual need you had when you asked the question.

Common Prompting Mistakes

❌ Accepting the First Answer Blindly

People ask once, copy the output, and move on.

Problem: The output may be shallow, incorrect, or misaligned with what you actually need.

💡 AI can be **confidently wrong**. If you accept the first answer without questioning, you might spread misinformation. Great AI users **iterate** and **verify**.

❌ Overloading the Prompt

Users dump everything into one prompt, hoping AI will figure it out.

Example: "Explain robots, give examples, compare with humans, make it funny, short, detailed, and include a story."

Problem: Conflicting instructions ("short" vs "detailed"), AI tries to satisfy everything.

💡 Clarity > complexity. Break complex requests into **multiple, focused prompts** or use **iterative prompting** to build gradually.

Weak vs. Strong Prompts

Let's see how adding elements and avoiding mistakes transforms a weak prompt into a strong one.

Weak Prompt	Strong Prompt
"Make this better." (<i>assumes AI understands</i>)	"Rewrite this paragraph to be more formal and professional. Keep it the same length."
"Explain photosynthesis" (<i>incomplete information</i>)	"Explain how photosynthesis works to a 5th grader. Use simple words and include an analogy."
"Tell me about the Eiffel Tower." → No follow-up. (<i>no iteration</i>)	After first answer: "That's good. Now add three interesting facts that most people don't know."
"Write a funny story about a robot, but also make it educational, short, and include a poem." (<i>overloading</i>)	"Write a short, funny story about a robot. After that, I'll ask for an educational explanation separately."
"Give me a study guide for the water cycle." → (<i>not checking output</i>)	"Create a study guide for the water cycle. I will then check it against my textbook for accuracy."

Mini Activity 2 : Rewrite Weak Prompts

Instructions: For each weak prompt below, try adding helpful elements that make sense. Use some of the key elements we discussed (Clear Instructions, Context, Format, Constraints) if they help. There's no single correct answer - experiment and see what works.

#	Weak Prompt (with mistakes)	Your Stronger Prompt
1	"Tell me about the solar system"	
2	"Write a poem"	
3	"Give me a recipe"	
4	"Explain how a battery powers a flashlight"	
5	"Draw a superhero"	
6	"What's the weather?"	
7	"Help me plan a surprise party"	
8	"Give me ideas for a comic strip about a penguin who learns to fly"	

1 Solar System

Describe the solar system's planets in order from the sun. I'm a 4th grader doing a poster. Use a numbered list with one sentence per planet. Include the planet's distance from the sun and one fun fact.

3 Recipe

Give me a simple recipe for chocolate chip cookies. I'm a 5th grader baking for the first time. Numbered steps. Use only ingredients commonly found at home; no nuts.

5 Superhero

Draw a cartoon superhero who controls plants. For a comic book aimed at kids ages 8–10. Describe the character's appearance, costume, and powers. Make the character friendly, not scary.

7 Surprise Party

Help me brainstorm ideas for a surprise party. For my best friend's 12th birthday. They love dogs and video games. Bulleted list of 5 party ideas. Budget under \$100; keep it a secret until the day.

2 Poem

Write a short poem about a rainy day for a school poetry contest. 4 lines with 6-8 words each. Use simple words; mention rain and an umbrella.

4 Battery & Flashlight

Explain how a battery powers a flashlight. I'm an 8th grader studying circuits. Use a diagram described in words. Keep it under 100 words; include the terms "circuit" and "electrons."

6 Weather

Tell me today's weather forecast. I'm planning a picnic outdoors. Short summary with temperature, chance of rain, and wind speed. Only include information relevant to outdoor activities.

8 Penguin Comic Strip

Help me brainstorm ideas for a comic strip about a penguin who learns to fly. I'm creating it for fun, for kids ages 7-10. A list of 5 plot ideas, each with 2 sentences. Keep it funny and positive. No sad endings.

Iterative Prompting

Iterative prompting is a technique where AI interactions are refined through multiple rounds of feedback rather than a single attempt.

Write Your Prompt

Start with your best attempt using all four components.

Refine and Repeat

Adjust your prompt based on what you learned and try again.



Get AI Response

Submit your prompt and receive the AI's output.

Evaluate the Output

Does it match what you needed?
What's missing or wrong?

Iterative Prompting in Action

Example: Building a story about a lost cat through three iterations.



Prompt 1

"Write a story about a lost cat."

Response: A very short, simple story.

Prompt 2

"Write a longer story about a lost cat. Make it adventurous and include a friendly dog who helps."

Response: Better, but still missing a setting.

Prompt 3

"Write a 3-paragraph story about a lost cat named Whiskers who gets help from a friendly dog in a busy city park. Make it exciting and end with them becoming friends."

Response: Perfect!

The Prompt Framework

To make it easy to remember helpful elements, you can use this simple framework. It's a way to combine Clear Instructions, Context, Format, and Constraints into one clear instruction.

[Action] + [Topic] + [Context] + [Format] + [Constraints]

Action

What should the AI do? (e.g., list, explain, create)

Topic

What is it about? (e.g., solar system, a recipe)

Context

Who is it for? Why? (e.g., for a 4th grader)

Format

How should it look? (e.g., bullet points, 2 paragraphs)

Constraints

Any limits? (e.g., keep it short, use simple words)

The Prompt Framework - Example

Action

Explain

Topic

how to care for a pet hamster

Context

I'm a 5th grader who has never owned a pet before

Format

a checklist of daily, weekly, and monthly tasks

Constraints

Keep it simple; no more than 10 items

Final prompt: "Explain how to care for a pet hamster. I'm a 5th grader who has never owned a pet before. Give me a checklist of daily, weekly, and monthly tasks. Keep it simple and no more than 10 items."

How the Framework Helps Avoid Common Mistakes

Common Mistake	How the Framework Helps
Assuming AI understands you	You specify the action clearly.
Giving incomplete info	You provide context.
Accepting first answer blindly	You'll iterate, not accept blindly.
Overloading the prompt	The framework keeps it clear and structured.
Not checking output	You'll evaluate against your goal.

 **Practice with yourself:** For any task, you can use as many or as few of these elements as you need.



Hands-On Activity: Prompt Playground

Goal: Play with prompts! Experiment with different elements, see how changes affect the AI's output, and start building your Prompt Library. There's no right or wrong - just explore.

Setup: Use ChatGPT or Gemini.

Prompt Playground: Three Tasks

Task 1: Curiosity Play

Choose something you're curious about (e.g., How do birds know where to fly in winter?). Start with a simple prompt, then add one helpful element at a time. Write down how each change improves the answer or if it doesn't, that's okay too!

Task 2: Creative Play

Ask AI to help you create something fun - a character, a short story, a game idea. Try at least two different formats (e.g., bulleted list vs. dialogue). Compare the results. Which one worked better for your goal?

Task 3: Real-Life Play

Use the framework (Action + Topic + Context + Format + Constraints) if it helps. Solve a real problem (e.g., plan a weekend, write a kind message, learn a skill). Run it, then refine once.

Summary

Key Term	What It Means
Prompt	Your instruction to AI
Iterative	Improving through repeated attempts
Constraints	Limits you give to AI (length, style, what to avoid)

Summary

Helpful Elements to Consider (Use as Needed)

- Clear Instructions - tell AI exactly what to do (include action and topic)
- Context - who you are, what you need
- Format - how you want the answer
- Constraints - limits (length, style, what to avoid)
- Optional Example - show the AI what you want

Common Mistakes to Avoid

- Assuming AI understands you
- Giving incomplete information
- Accepting the first answer blindly
- Overloading the prompt
- Not checking output against need

 **The Prompt Framework** : Action + Topic + Context + Format + Constraints

The Most Important Idea

There's no single "perfect" way to write a prompt.

The best prompt is the one that helps you get what you need. Use helpful elements when they make sense, iterate when you're not satisfied, and always stay in control of the final output. This skill will serve you in every AI interaction - in this course, in school, and in the future.

Homework :

My Prompt Library

Your Prompt Library will be a **living document** you keep throughout the course and beyond. It's where you'll save your best prompts, track what worked, and quickly find the perfect instruction for any task. This structure is designed to grow with you as AI evolves. You decide what to include - some entries might be simple, others more detailed.

Create a document titled: "**My Prompt Library**". Use the following structure for each entry. For this first assignment, add at least two entries from today's activity.

- ❏ **Assignment:** Add at least two entries from today's activities. Keep your library organized in a way that works for you. As you add more prompts in future sessions, you'll build a powerful reference that will save you time and help you become a confident prompt engineer.



Prompt Library Entry Format (Flexible)

Field	What to Write (optional fields are marked)
Date	When you used this prompt
Goal	What you wanted to achieve (e.g., "Plan a weekend trip," "Explain how a battery works")
AI Tool	Which tool you used (ChatGPT, Gemini, etc.)
Final Prompt	The exact prompt you used (the best version)
Elements Used	Optional: Which helpful elements you included (Clear Instructions, Context, Format, Constraints, Example)
What Worked	Optional: Why this prompt gave you a good result
What I'd Change Next Time	Optional: What you learned for future prompts
AI's Response	Optional: Copy the response (or a summary)

Prompt Library: Example Entry

Field	Example
Date	March 26, 2026
Goal	Get simple ideas for a poster about the water cycle
AI Tool	ChatGPT
Final Prompt	Give me ideas for a water cycle poster for 6th graders. Include 3 main stages with simple diagrams described in words. Use bullet points.
Elements Used	Clear Instructions (give ideas), Context (6th graders), Format (bullet points), Constraints (3 stages, simple diagrams)
What Worked	Being specific about the grade level made the language perfect for my audience.
What I'd Change Next Time	I could ask for real-world examples of each stage to make the poster more engaging.
AI's Response	• Evaporation: Show the sun heating a puddle, with arrows pointing upward...

Next Session Preview

Session 12: Advanced Prompting Techniques

Now that you've mastered the basics and learned to avoid common mistakes, we'll level up:



Few-shot prompting

Teaching AI by giving examples.



Chain-of-thought

Asking AI to explain its reasoning.



Role-playing

Asking AI to act as a specific persona.

Your Prompt Portfolio will grow with even more powerful examples